INCH-POUND

MIL-DTL-32015/2B(AS)
6 May 2004
SUPERSEDING
MIL-DTL-32015/2A(AS)
22 May 2000

DETAIL SPECIFICATION SHEET

MICROPHONE, LINEAR, MBU-12, -20P AND –23/P OXYGEN MASKS M-176/AIC

This specification is approved for use by the Naval Air Systems Command, Department of Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32015.

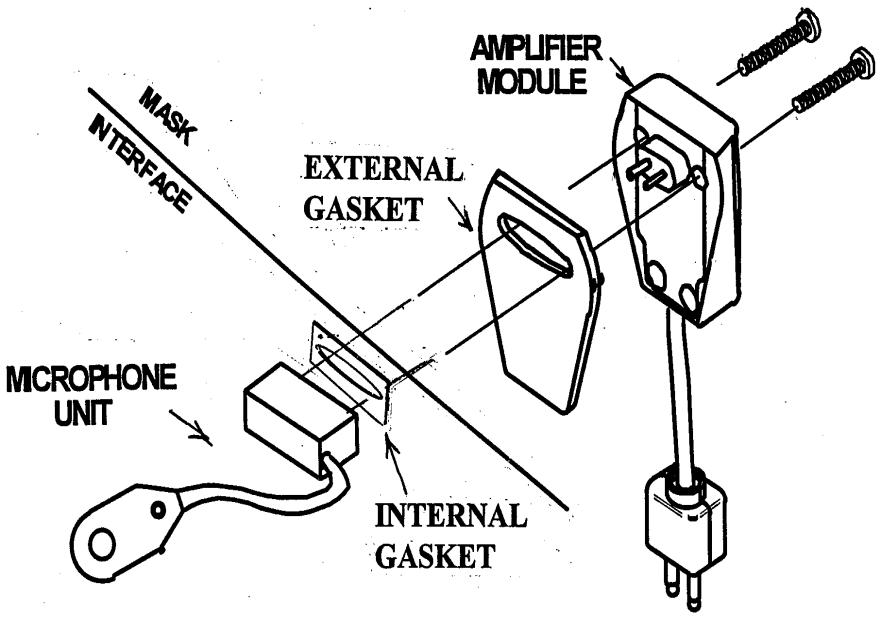


FIGURE 1. Amplifier w/microphone assembly, M32015/2-1 (exploded view).

AMSC N/A FSC 5965

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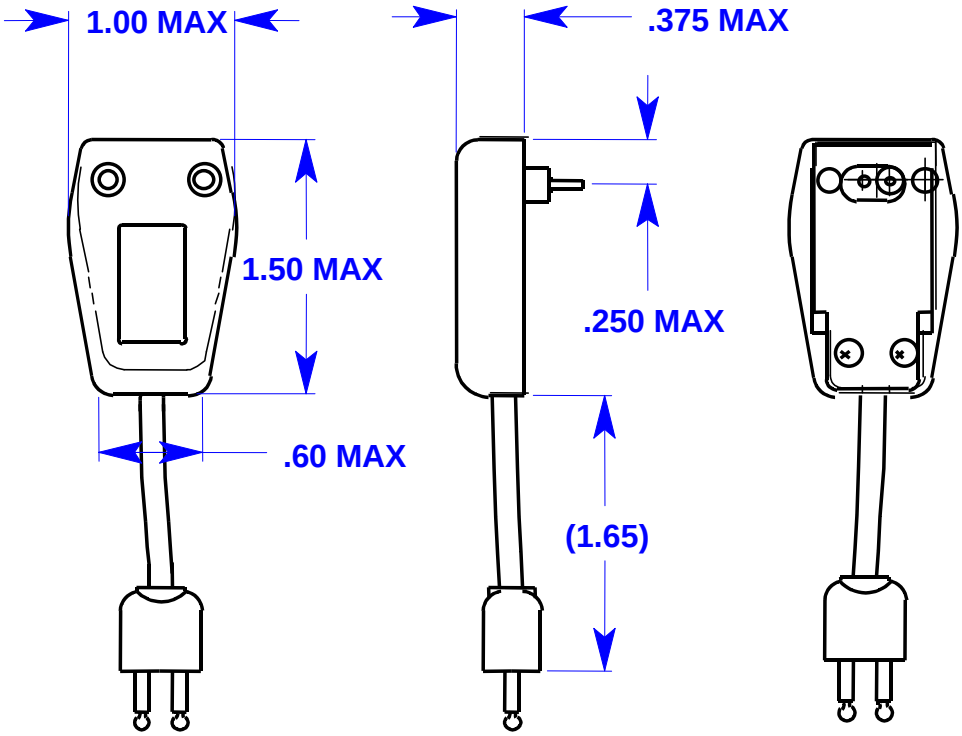


FIGURE 2. Amplifier assembly.

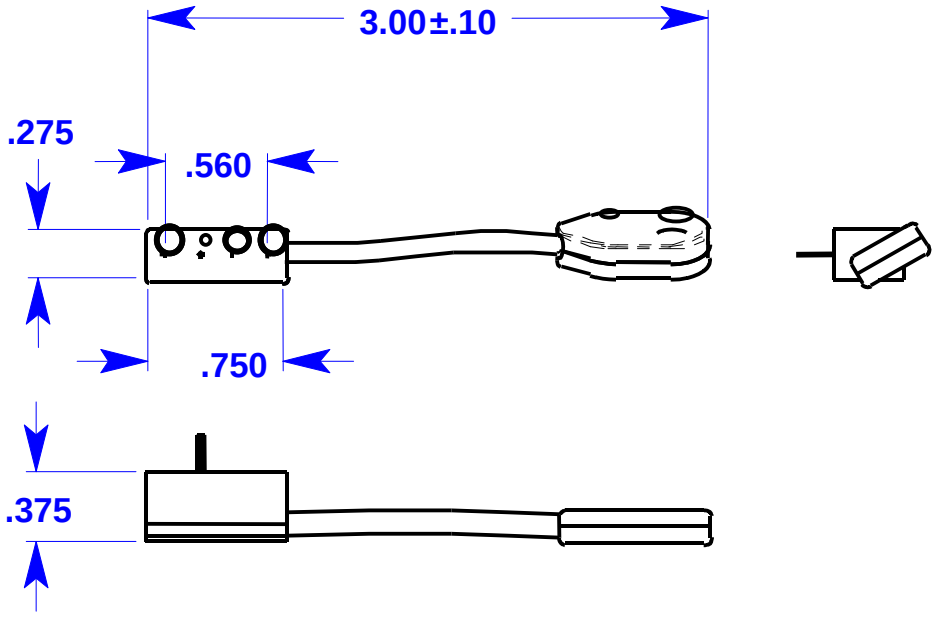


FIGURE 3. Microphone unit,

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APPLICABLE DOCUMENTS:

NASM35275	Screw, Machine-Drilled Fillister Head, Slotted, Corrosion Resisting Steel, Passivated, UNC-2A

(Application for copies of NASC publications should be addressed to the Aerospace Industries Association, 1250 Eye Street NW, Washington DC 20005, or matthew@aia-aerospace.org.)

APPLICABLE DRAWINGS:

AIR FORCE

57B12662	Connector, Plug Electrical U-173/U
57B12664	Tip-Electrical, Plug Connector

(Copies of the above drawings are available from Defense Supply Center Columbus, DSCC-VTCD, P.O. Box 3990, Columbus, OH 43216-500, or www.charles,mcgrath@dla.mil.)

REQUIREMENTS:

DESIGN and CONSTRUCTION: See Figures1 through 3.

*MICROPHONE/ AMPLIFIER INTERFACE: The microphone amplifier interface at the oxygen mask surface shall seal to protect the electrical contacts from severe moisture effects inside the oxygen mask and to prevent oxygen from the mask. The internal gasket shall be permanently attached to the mounting bracket.

CONFIGURATION: The microphone assembly is comprised of a preamplifier assembly and a microphone element assembly. The microphone element assembly shall contain the microphone element and only the electronics necessary for transfer of the audio signal to the preamplifier. All electronics shall be contained within the preamplifier assembly rather than the microphone element assembly to minimize the risk of a fire potential within the oxygen mask. Voltages and currents in the microphone element assembly shall not create sparks or any other adverse effect which presents a fire potential inside the oxygen mask.

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COMPONENT PARTS: The microphone assembly shall be supplied with the following:

1 Amplifier assembly (see figure 2).
1 External gasket which shall be attached to the bottom of the amplifier via adhesive.
1 Internal gasket.
1 Microphone element (see figure 3).
2 NASM35275-210 machine screws, fillister head (2-56 x 3/4 lg.).

PART or IDENTIFYING NUMBERS (PIN): M32015/2-1

WEIGHT: 62g max. for the assembly.

COLOR: Semigloss, lusterless, black.

CONNECTOR:

OUTPUT CONNECTOR: The output connector shall contain two pins made in accordance with Air Force Drawing 57B12664. The pins shall be arranged as detailed in Air Force Drawing 57B12662 and together shall comprise the U-173/U connector.

PULL AND TWIST: The connector or amplifier shall not part from the conductor at the specified loads after twisting and pulling as specified below. The twist test shall be performed before the pull test on the same amplifier assembly. There shall be no more than 1/32 inch slippage of the cable jacket from the connector or amplifier.

TWIST TEST: Each amplifier assembly shall be secured in a holding device in so that it shall not rotate. Gripping the connector, the twist shall be applied about the axis of the cable to cause a point on the cable to twist 180 degrees from the starting position and be held there for 30 seconds. The cable shall then be twisted in the opposite direction such that the point on the cable has twisted 180 degrees past the original starting position and be held there for 30 seconds.

PULL TEST: The amplifier assembly to be tested shall be secured in a stationary holding device and the connector gripped. A pull force of 20 ± 2 foot pounds shall be applied for 30 seconds.

AMPLIFIER TO MICROPHONE PINS: The pin configuration shown in figures 2 and 3 are for information only since the pins are within the system design and not required to be compatible with external connectors. The connector design shall be within the envelope dimensions shown for compatibility with the oxygen mask design.

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AMPLIFIER AND MICROPHONE SUBASSEMBLIES: The amplifier and microphone subassemblies shall be constructed such, when assembled in the oxygen mask with the MS 35275-210 screws, they will not be damaged when 12 ± 3 in-oz. of torque is applied to the screws.

AMPLIFIER TO MICROPHONE SEAL: The amplifier to microphone seal shall be constructed such, when assembled in the oxygen mask with the NASM35275-210, screws they will create an air tight seal. The configuration of the internal and external gaskets shown on figure 1 is for reference only, the actual shape shall match the amplifier and microphone components.

INPUT CURRENT: 20 mA max. at 27.5 VDC when connected as shown on figure 5 with switch SW1 open.

IMPEDANCE: The electrical impedance of the microphone assembly shall be such that the unit shall meet the sensitivity and frequency response to a close sound source while connected to a 100 ohm ±1 percent load impedance.

*SENSITIVITY: 64 mV to 130mV maximum 104 dB SPL with 27.5 VDC supply voltage, and shall not vary by more than 1.0 dB when the supply voltage is varied between 22 and 30 volts.

SENSITIVITY AT REDUCED BIAS VOLTAGE: Shall not vary by more than 12 dB at 104 dB SPL from the 27.5 VDC established value when the supply voltage is varied between 12 and 22 volts.

SENSITIVITY AT ALTITUDE: ±3 dB of sensitivity at ground level while at a simulated altitude of 15,000 ft above sea level.

FREQUENCY RESPONSE AT GROUND LEVEL: See figure 6.

FREQUENCY RESPONSE AT ALTITUDE: Within ±5 dB of the frequency response at sea level while at a simulated altitude of 15,000 feet above sea level.

NOISE IMMUNITY: See figure 7.

*LINEARITY: The microphone output level shall be a linear function of acoustic input sound pressure level (SPL) from 74 to 120 dB, when connected in accordance with figure 5.

HARMONIC DISTORTION: As specified in the general specification.

GAIN STABILIZATION: Within 100 ms when the supply voltage is between 23 and 30 VDC. Within 150 ms when supply voltage is between 12 and 22 VDC.

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OVERVOLTAGE PROTECTION: 15 minutes operation in the test circuit shown on figure 5 with 27.5 volts applied while the 330 ohm resistor is shorted.

EXPLOSIVE ATMOSPHERE: As specified in the general specification.

EXPLOSIVE DECOMPRESSION: As specified in the general specification

ENVIRONMENTAL STRESS SCREENING (ESS): 24 hours of temperature cycling between -20 and +85 ºC, with each cycle consisting of 1 hour at each temperature and the change between temperature occurring at a minimum rate of 5 ºC per minute. The microphone assemblies shall be functionally tested prior to each cold to hot transition and upon completion of the temperature cycling.

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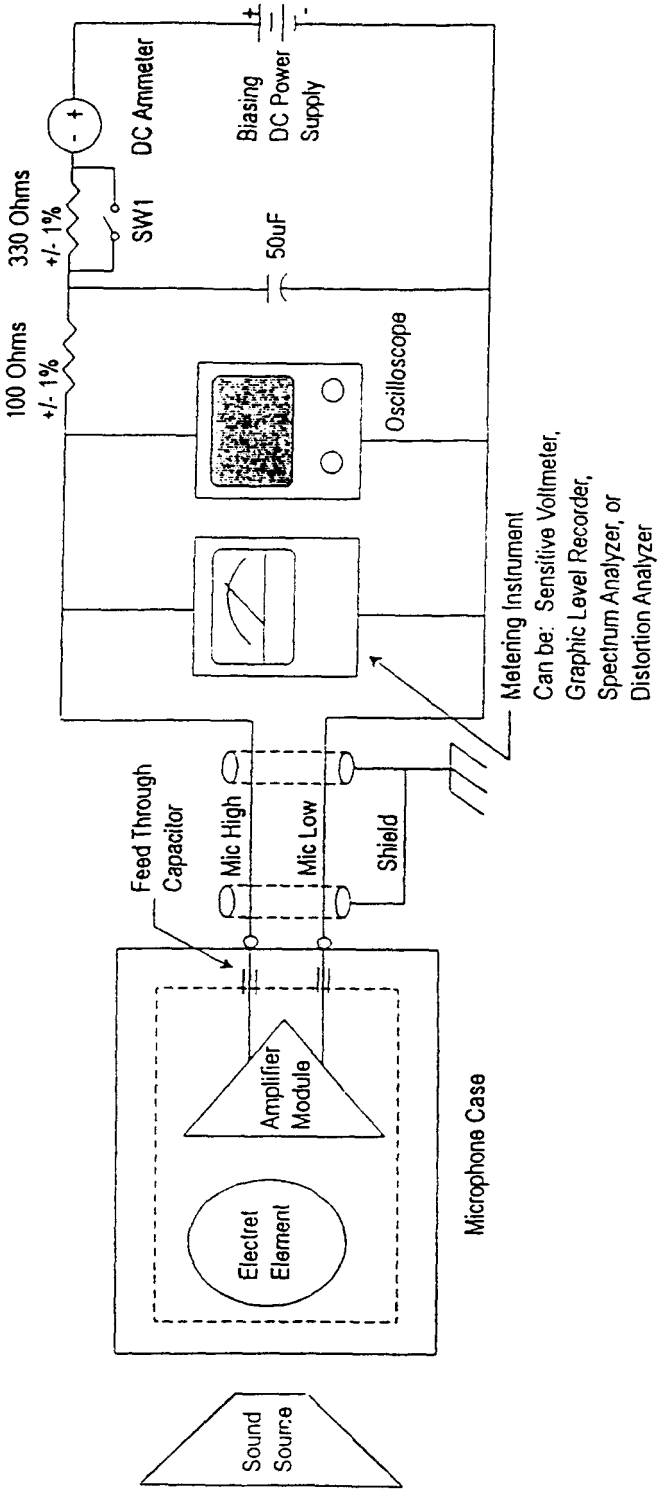


FIGURE 5. Test circuit schematic.

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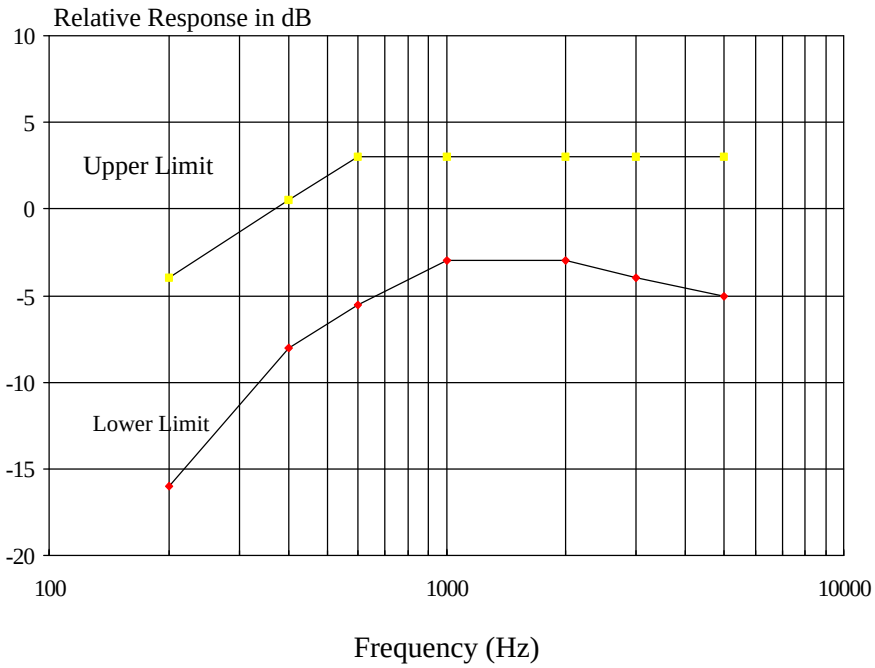


FIGURE 6. Frequency response limit curves.

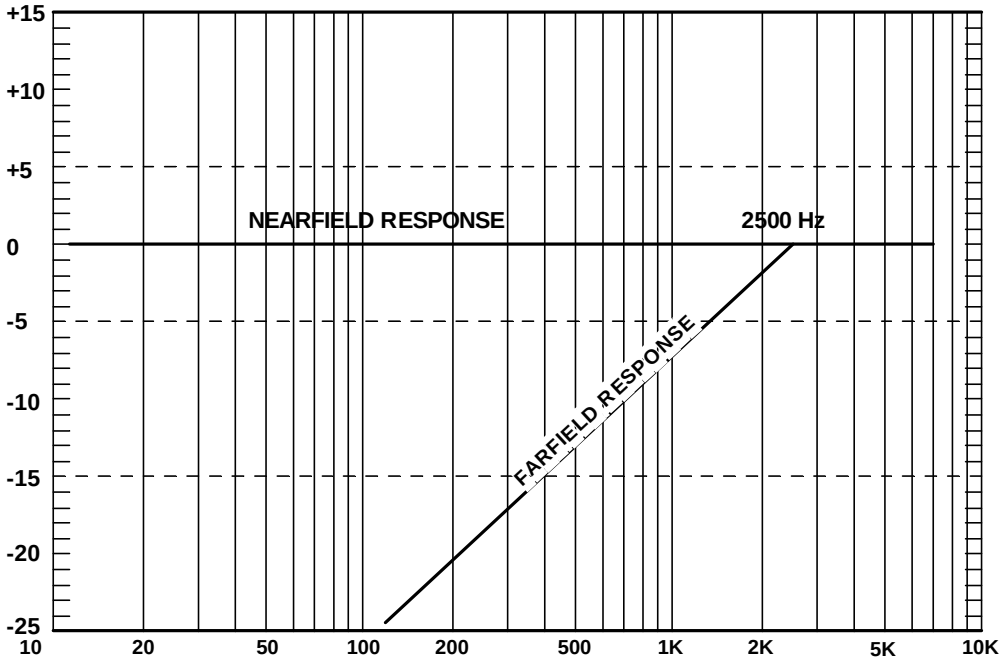


FIGURE 7. Noise immunity characteristics.

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MARKING:

M-176/AIC
M32015/2-1
MANUFACTURER'S NAME OR CAGE CODE
CONTRACT NUMBER
DATE CODE
SERIAL NUMBER

INTENDED USE: The microphone, M32015/2-1, is a noise-canceling, DC-powered, high-gain microphone designed for use in a pressure-type oxygen mask. The microphone is intended for use as a part of the airborne communication system which will provide communication of high intelligibility under the extreme noise conditions encountered in military aircraft. The microphone is designed to work with the AIC-14 intercommunication system, specifically control box C2645D.

NOTE: The margins of this amendment are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
Navy - AS
(Project 5965-0389)

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